MUH401- INTERIM PROJECT REPORT FORMAT

Cover Page

Title of the Project:	**Machine Learning-Based Discrimination of Precipitation-Induced Radon Washout for Radiological Anomaly Detection in EPA RadNet Air Monitoring Network**
Course Name:	MUH401
Department / Faculty:	Department of Environmental Engineering / Faculty of Engineering
University:	Hacettepe University
Student Name(s) & ID(s):	Necdet Ömer Barut (2230867059)
Supervisor / Instructor:	Prof. Dr. Şule Ergün
Submission Date:	November 2026
Academic Year / Semester:	2026–2027 Fall Semester

ABSTRACT

This report presents a gradient-boosted decision tree framework using physics-motivated spectral and meteorological features to suppress precipitation-induced radon progeny washout false alarms in automated radiation monitoring networks. Scope: detection performance is evaluated primarily on synthetic radiological plumes added onto real United States Environmental Protection Agency (EPA) RadNet background measurements; false alarms are counted on the real, uninjected background. Real radiological release feasibility is benchmarked on historical 2011 trans-Pacific observations. During rainfall events, natural terrestrial radon daughters ^{214}Pb and ^{214}Bi are scavenged onto particulate air filters. Across 15,602 observed synchronous rain hours in this study, precipitation induces a median gross count change of $+6.9\%$ (mean $+12.05\%$), with the 90th percentile reaching $+41.4\%$ and convective thunderstorm peaks surging up to $+196.7\%$, while coastal stations experience slight count depressions (-8.7% median in San Diego). On the 2023–2025 test dataset (106,628 observed hours = 12.16 station-years), conventional local rolling 7-day 3σ thresholds trigger 82.62 false alarm episodes per station-year pooled across the five-station network (66.37 to 96.17 per station; global dry 3σ triggers 52.86 pooled, 9.44 to 78.64 per station), while over the full 2017–2025 period the rolling 7-day mean plus $3\sigma_{\text{dry}}$ rule triggers 6.59 to 62.91 false alarms per station-year across all five stations (52.98 to 62.91 across Birmingham, Washington, and Dallas, where over 69% of episodes coincide with rainfall). The project evaluates 334,567 valid RadNet hours (324,550 synchronous radiation-weather hours) across five stations spanning 2017 through 2025: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL. Detection performance is benchmarked across 450 filter-synchronized synthetic release plumes covering ^{137}Cs, ^{131}I, ^{134}Cs, ^{60}Co, and ^{132}Te/^{132}I. On the held-out 2023–2025 test dataset, the 59-feature

model (`Tier 4`) achieves 2.71 ± 1.39 false alarm episodes per station-year with a 95% station-month block bootstrap confidence interval of [1.17, 4.58] and $94.00 \pm 0.93\%$ synthetic plume detection at the 95% validation target. In leave-one-station-out cross-validation, 4 of 5 stations maintain ≤ 1.40 false alarms per station-year, whereas unadapted transfer to Tampa yields 169.42 false alarms per station-year. Feature importance analysis shows that elapsed time since rain and local spectral stripping residuals dominate model split gain, while Bateman washout ODE and dose-consistency residuals contribute smaller secondary gains.

Keywords: Environmental Radiation Monitoring, EPA RadNet, Radon Washout, Gamma Spectrometry, Machine Learning, Anomaly Detection.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CI	Confidence Interval
CPM	Counts Per Minute
eFDR	Empirical False Discovery Rate
EPA	United States Environmental Protection Agency
FA/yr	False Alarms per Station-Year
FWHM	Full Width at Half Maximum
GBDT	Gradient Boosted Decision Tree
ISD	Integrated Surface Dataset
LOSO	Leave-One-Station-Out
MLP	Multi-Layer Perceptron
NaI(Tl)	Thallium-activated Sodium Iodide Scintillation Detector
NCEI	National Centers for Environmental Information
NOAA	National Oceanic and Atmospheric Administration
NORM	Naturally Occurring Radioactive Material
OOD	Out-of-Distribution
ROC	Receiver Operating Characteristic
ROI	Region of Interest
UTC	Coordinated Universal Time
WMO	World Meteorological Organization

1. INTRODUCTION

1.1 Background of the Problem

Continuous atmospheric radiation surveillance networks are operated worldwide by regulatory authorities to provide immediate detection of clandestine nuclear testing, industrial accidents, and radiological dispersion events (Sangiorgi et al., 2020). Large-scale airborne monitoring was demonstrated during the transboundary tracking of radionuclide plumes from the Fukushima Dai-ichi incident (Masson et al., 2011). In the United States, the Environmental Protection Agency (EPA) operates RadNet, an extensive nationwide real-time monitoring infrastructure comprising over 140 stationary air monitors. Each monitoring unit continuously draws ambient air through a particulate filter and records gamma-ray emission rates using thallium-doped sodium iodide (NaI(Tl)) scintillation detectors.

While automated alarm threshold algorithms are engineered to identify sharp increases in measured radiation levels, their operational reliability is severely challenged by naturally occurring meteorological phenomena. Atmospheric precipitation scavenges airborne daughter radionuclides of radon-222 (^{222}Rn), primarily lead-214 (^{214}Pb) and bismuth-214 (^{214}Bi), depositing them directly onto the ground surface and particulate filter media (Minato, 1980; Burnett et al., 2010). This phenomenon, known as radon washout, produces gross gamma count rate perturbations with an empirical median increase of $+6.9\%$, with the 90th percentile reaching $+41.4\%$ and severe convective thunderstorms surging up to $+196.7\%$ above baseline (Hsieh et al., 2025). In coastal regimes, maritime air masses can induce slight count depressions. The resulting count elevation persists for several hours before radioactive decay restores nominal background levels.

1.2 Problem Statement

Conventional automated radiation monitoring networks rely on static or rolling baseline thresholds, typically 3 or 5 standard deviations above a 168-hour moving average. Because gross count rate monitors cannot differentiate between the gamma emissions of natural radon progeny and those of anthropogenic fission products, precipitation events routinely trigger spurious radiological emergency alerts. Across the five-station network on the 2023–2025 test split (12.16 station-years), local rolling 7-day 3σ threshold algorithms produce 82.62 false alarm episodes per station-year pooled (ranging per station from 66.37 in San Diego to 82.69 in Dallas, 86.22 in Washington, DC, 86.54 in Birmingham, and 96.17 in Tampa), while global dry 3σ produces 52.86 pooled (9.44 to 78.64 per station). Over the full 2017–2025 evaluation of the rolling 7-day mean plus $3\sigma_{\text{dry}}$ rule (`rolling_cpm_3s`), per-station alarm rates are 6.59 in Tampa, 20.91 in San Diego, 52.98 in Washington, DC, 53.57 in Birmingham, and 62.91 in Dallas, with over 69% of episodes in Birmingham, Washington, DC, and Dallas coinciding with rainfall within 3 hours.

Conversely, raising the alert threshold to conservative levels such as 5σ suppresses false alarms to 23.51 per station-year on the 2023–2025 test split but severely degrades synthetic plume detection capability to 6.00% (median delay 8.0 h) under classical counting limits (Currie, 1968). Stronger non-ML and unsupervised multi-channel baselines will be evaluated in .

1.3 Importance and Relevance of the Study

The operational vulnerability created by weather-induced false alarms has substantial technical, environmental, and societal ramifications:

- **Operational Integrity and Alarm Fatigue:** Frequent false alarms desensitize emergency personnel, increasing the hazard that a genuine radiological incident may be dismissed as a weather-

induced anomaly.

- **Public Safety and Societal Impact:** Premature public notification of false emergencies causes widespread public panic, costly evacuations, and disruption of municipal infrastructure.
- **Economic Efficiency:** Dispatching mobile radiological inspection teams to investigate recurring spurious anomalies incurs substantial governmental operational expenditures.
- **Scientific Contribution:** By integrating multi-channel scintillation spectrometry with multi-scale atmospheric thermodynamic features using gradient-boosted decision trees on physics-motivated features, and drawing context from domain-informed nuclear safety and transport modeling (Lye et al., 2025; Ayoub et al., 2024a), this work benchmarks multi-modal washout discrimination on EPA RadNet data.

1.4 Clear Aim and Objectives of the Project

The primary aim of this project is to develop, validate, and benchmark a machine learning framework using physics-motivated features that discriminates precipitation-induced radon washout spikes from synthetic anthropogenic radiological anomalies in continuous environmental surveillance networks. Specific technical objectives include:

1. Harmonize longitudinal multi-channel gamma spectrometry from EPA RadNet and contemporaneous surface meteorology from the NOAA Integrated Surface Dataset (Smith et al., 2011) across five climatologically distinct monitoring stations spanning nine full calendar years from 2017 through 2025.
2. Engineer a strictly causal real-time feature space that captures differential gamma spectral peak ratios and antecedent precipitation dynamics without future lookahead bias.
3. Formulate a multi-tier machine learning architecture to continuously categorize observations into dry background, radon washout, or synthetic anthropogenic release.
4. Conduct a 20-seed benchmark against conventional statistical baselines and rain-masked control rules over 450 synthetic radiological releases to evaluate false alarm rates and synthetic detection sensitivity.

1.5 Scope and Boundaries of the Study

The scope of this project encompasses five EPA RadNet monitoring stations selected across distinct Köppen climate zones: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL. The temporal boundary spans January 1, 2017, through December 31, 2025, providing 334,567 valid RadNet hours and 324,550 synchronous radiation-weather station-hours.

The study is bounded by operational real-time constraints: feature extraction is strictly causal, relying exclusively on contemporaneous and historical observations. Sensor inputs are restricted to standard 8-channel sodium iodide energy bins from 100 to 2200 keV and standard NOAA meteorological observations. Source releases are evaluated primarily through 450 synthetic plume injections constructed with parametric deposition rate curves (`linear_ramp`, `step`, `sigmoidal`, `exponential`) and single- and two-step Bateman particulate filter accumulation/decay kinetics, covering ^{137}Cs, ^{131}I, ^{134}Cs, ^{60}Co, and ^{132}Te/^{132}I mixtures motivated by historical reactor release inventories (Stohl et al., 2012). No

atmospheric dispersion model is executed in this repository. Historical release observations are evaluated separately in the 2011 real-event analysis. Detection metrics reflect performance conditional on these spectral templates and physical deposition dynamics.

2. THEORETICAL BACKGROUND / LITERATURE REVIEW

2.1 Key Theories, Models, Concepts, and Definitions

Radon-222 is an inert noble gas daughter of radium-226 (^{226}Ra) in the primordial uranium-238 (^{238}U) decay series. Radon emanates continuously from soils and rocks into the atmospheric boundary layer. Because radon has a half-life of 3.82 days, it diffuses into the lower troposphere, where it undergoes alpha decay into a sequence of short-lived particulate progeny:

$$^{222}\text{Rn} \xrightarrow[\alpha]{3.82\ \text{d}} {}^{218}\text{Po} \xrightarrow[\alpha]{3.10\ \text{m}} {}^{214}\text{Pb} \xrightarrow[\beta^-]{26.8\ \text{m}} {}^{214}\text{Bi} \xrightarrow[\beta^-]{19.9\ \text{m}} {}^{214}\text{Po} \xrightarrow[\alpha]{164.3\ \mu\text{s}} {}^{210}\text{Pb} \tag{1}$$

Lead-214 (^{214}Pb) and bismuth-214 (^{214}Bi) rapidly attach to submicron ambient aerosol particles. During rainfall events, these radioactive aerosols are captured via in-cloud rainout and below-cloud washout (Takeuchi and Katase, 1982; Paatero, 2000; Sportisse, 2007). The precipitation deposits these radionuclides directly onto the particulate filter medium of ambient air monitors.

Because ^{214}Pb emits prominent gamma lines at 295.2 and 351.9 keV, and ^{214}Bi emits high-energy gammas at 609.3, 1120.3, and 1764.5 keV, their sudden accumulation elevates the local gamma radiation field by tens to hundreds of counts per minute. Following the cessation of rain, the activity decay curve of the deposited progeny is governed by the effective half-life of the ^{214}Pb/^{214}Bi decay chain, returning to nominal baseline within 2 to 4 hours (Mercier et al., 2009; Melintescu et al., 2018).

2.2 Relevant Technical Background: Scintillation Spectrometry

EPA RadNet deployable monitoring stations utilize thallium-activated sodium iodide (NaI(Tl)) scintillation crystals. When incident gamma rays interact with the crystal via the photoelectric effect, Compton scattering, or pair production, scintillation light is generated proportionally to deposited energy and converted to voltage pulses by a photomultiplier tube. While sodium iodide detectors exhibit modest energy resolution of approximately 7% full width at half maximum at 662 keV, they are rugged, thermally stable, and well-suited for long-term field deployment (Knoll, 2010). However, outdoor ambient temperature variations can induce slight photopeak drift and light-yield shifts (Moszyński et al., 2006; Wen and Zhou, 2024), underscoring the importance of robust energy channel grouping and spectral normalisation.

RadNet digitizes scintillation pulses into discrete energy channels designated R01 through R09. Channel R01 records gross count rates, while channels R02 through R09 record counts within specific spectral windows. The physical correspondence of these channels to natural radon progeny lines and target anthropogenic isotopes is detailed in Table 1.

Table 1: EPA RadNet Discrete Gamma Energy Channel Boundaries

Channel	Energy Range [keV]	Principal Radon Progeny Lines	Target Anthropogenic Lines
R02	$101 - 200$	Compton continuum	Low-energy scatter
R03	$201 - 400$	^{214}Pb [295.2, 351.9 keV]	^{131}I [364.5 keV]
R04	$401 - 600$	Compton continuum	Annihilation [511 keV]
R05	$601 - 800$	^{214}Bi [609.3 keV]	^{137}Cs [661.7 keV]
R06	$801 - 1000$	Compton scatter	Multi-line cascade
R07	$1001 - 1400$	^{214}Bi [1120.3 keV]	^{60}Co [1173.2, 1332.5 keV], ^{134}Cs
R08	$1401 - 1800$	^{214}Bi [1764.5 keV], ^{40}K [1460.8 keV]	Terrestrial background
R09	$1801 - 2200$	High-energy background	Cosmic rays and actinide daughters

2.3 Review of Relevant Previous Studies

The challenge of suppressing meteorological radon interference in environmental monitoring networks has been addressed through four distinct research paradigms:

1. **Dose Rate Time-Series Anomaly Detection:** Classical networks monitor gross count deviations against moving baselines (Burnett et al., 2010; Melintescu et al., 2018). More recently, Breitkreutz et al. (2023) developed neural network models to identify and quantify anomalies in environmental gamma dose rate time series, while Frankemölle et al. (2025) formulated Bayesian predictive methods to forecast local background radiation levels. While effective for scalar dose rate tracking, these approaches do not exploit multi-channel spectral features.

2. **Meteorological Scavenging and Atmospheric Coupling:** Radon washout spikes correlate strongly with precipitation dynamics and air mass history (Mercier et al., 2009; Paatero, 2000; Takeuchi and Katase, 1982). High-resolution minute-level analyses demonstrate that convective rain events produce sharp, non-linear radiation responses (Hsieh et al., 2025). Concurrently, meteorological data have been coupled with machine learning and neural operators to forecast atmospheric plume transport and contamination (Ayoub et al., 2024a, 2024b).

3. **Spectral Deconvolution and Nuisance Rejection:** To overcome gross count limitations, spectral analysis isolates characteristic isotope photopeaks. Meng and Ramsden (2000) evaluated spectral deconvolution methods for resolving overlapping gamma signatures. Detwiler et al. (2015) introduced spectral anomaly algorithms using natural terrestrial KUT nuisance rejection. Bilton et al. (2019) demonstrated non-negative matrix factorization (NMF) of gamma-ray spectra for robust background modeling and source discrimination.

4. **Machine Learning for Gamma Spectral Anomaly Detection:** Machine learning models have been applied to multi-channel gamma spectrometry for anomaly detection and radioisotope identification. Bandstra et al. (2023) explored explainable AI techniques for gamma-ray detection models. Alamaniotis (2024) developed mode-driven explainable AI for estimating background spectra in nuclear security. Abdelhakim (2025) integrated artificial neural networks with ant colony optimization for spectral anomaly detection, Karafasoulis and Kyriakis (2026) applied deep autoencoders for spectral anomaly detection, and Sun et al. (2024) designed self-attention neural networks for rapid multi-nuclide classification.

2.4 Gaps in Literature and Systematic Comparison

Existing literature exhibits three main limitations: first, gross count over-simplification, where operational networks discard multi-channel spectrometry and evaluate only total counts; second, synthetic benchmark limitations, where machine learning models rely on synthetic injection templates that may not account for particulate filter accumulation kinetics ; and third, weak comparator baselines, where complex models are rarely evaluated against meteorological masking rules and non-ML spectral projection baselines . This project evaluates multi-channel spectral ratios fused with multi-scale weather data against statistical and rain-masked baselines under validation-selected thresholds.

3. METHODOLOGY

3.1 System Description and Processing Pipeline

The computational pipeline developed in this project is illustrated in Figure 1. The framework ingests synchronous hourly observations from radiation and meteorological sensor networks, performs causal feature engineering, and feeds a gradient boosted decision tree classifier that assigns probabilities for three operational regimes: clean dry background, radon washout anomaly, and synthetic anthropogenic radiological release.

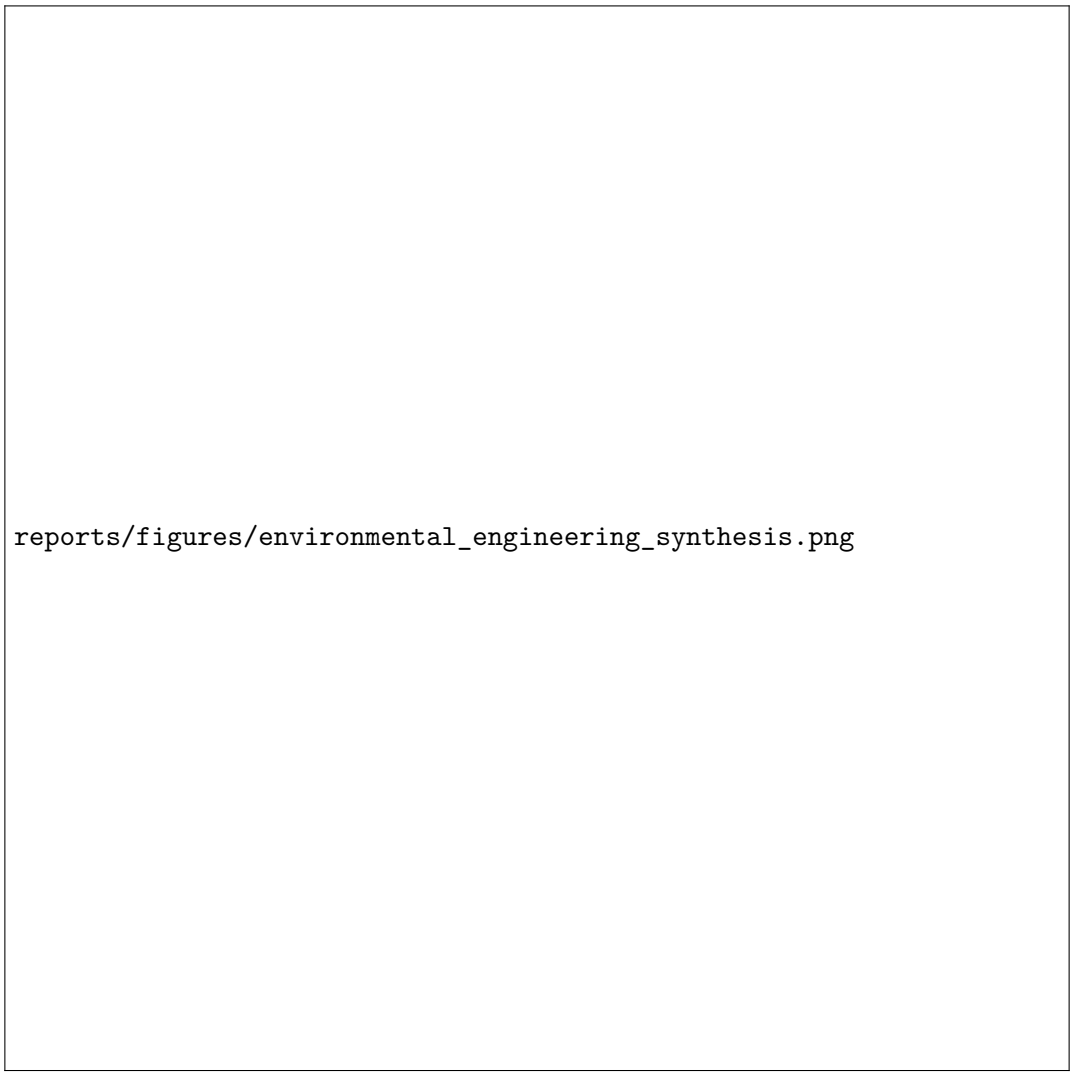


Figure 1: End-to-end environmental radiation anomaly detection and weather-fusion machine learning architecture using physics-motivated features.

3.2 Main Assumptions and Operational Boundaries

The computational architecture adheres to three operational boundaries and modeling assumptions:

1. **Strict Real-Time Causality:** To prevent data leakage, all rolling window statistics are computed using closed-left intervals $[t - W, t]$. No centered or forward-looking filters are permitted.

2. **Air Filter Accumulation Dynamics:** Airborne particulate monitors draw air through a glass-fiber filter that advances periodically. In 2021–2024 dry-spell step-drop analysis (intervals filtered to $[1.5, 14.0]$ days), detected filter advances occur with station median intervals of 2.92 to 4.54 days

and station mean intervals of 3.97 to 5.43 days, causing mean step-drops of 342.6 to 716.9 CPM (158 to 339 detected drops per station in 2021–2024; 357 to 1,295 over 2017–2025). An injected plume accumulates activity on the filter according to discrete convolution of the dimensionless deposition curve $c_{\text{dep}}(u)$ with radioactive decay:

$$A_{\text{filter}}(i) \propto \sum_{u=0}^{\min(i,t_{\text{passage}}-1)} c_{\text{dep}}(u)\, e^{-\lambda(i-u)} \tag{2}$$

which is normalized to unity at peak passage ($i = t_{\text{passage}} - 1$) and scaled directly by the sampled peak excess count rate (`peak_cpm`) without assuming an explicit volumetric flow rate Q. Multi-stage radioactive decay chains (^{132}Te $\rightarrow$ ^{132}I and ^{218}Po $\rightarrow$ ^{214}Pb $\rightarrow$ ^{214}Bi) are governed by analytical solutions of the Bateman equations (Cetnar, 2006).

3. **Assumed Detector Efficiencies and Dose Coupling:** Rather than Monte Carlo transport transfer factors , `src/nuclear_physics.py` uses a hard-coded 8-element vector of assumed 2"×2" NaI(Tl) intrinsic interaction efficiencies (`[0.92, 0.68, 0.48, 0.38, 0.32, 0.26, 0.21, 0.18]`) at channel centroid energies (0.15 to 2.00 MeV) combined with published ICRP Publication 74 ambient dose equivalent conversion coefficients $h^*(10)$ (ICRP, 1996), normalized to reproduce empirical train-only station radon washout regression slopes (`k_dose_channel`). Both the injected dose rate and the feature `dose_spectral_residual_z` currently share this `k_dose_channel` vector .

3.3 Input Data and Data Sources

Data ingestion leverages two primary US federal repositories:

- **EPA RadNet Air Monitoring Network:** Hourly gamma spectrometry records across 8 discrete energy bins from 100 to 2200 keV, ambient dose equivalent rate, and gross count rates. Data were retrieved via EPA public repositories spanning 2017 through 2025 across five stations: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL (334,567 valid RadNet hours).
- **NOAA Integrated Surface Dataset:** Contemporaneous hourly meteorological observations from colocated automated surface observing stations (Smith et al., 2011), yielding 324,550 synchronous radiation-weather hours. Extracted weather features (`WEATHER_FEATURES` in `src/feature_engineering.py`) comprise precipitation accumulation over 1, 3, 6, and 24 hours, binary rain indicators, elapsed hours since rainfall (≥ 0.1 mm), barometric pressure and 3-hour/24-hour pressure tendencies, dry-bulb temperature, dew point, and relative humidity.

3.4 Calculation Procedure and Causal Feature Engineering

Radiological and meteorological records are joined on a continuous UTC hourly calendar grid. While channel dry baselines (`channel_mu_dry`, `channel_sigma_dry`) and radon washout basis vectors (`radon_washout_basis`) are estimated strictly on the 2017–2020 fit fold, the scalar gross CPM dry normalizers (`mu`, `sigma`) use pre-test years 2017–2022. The causal feature engineering engine extracts 59 variables organized into four tiers:

1. **Gross Count Dynamics (Tier 1, 12 features):** Trailing 24-hour and 168-hour z-scores and

ratios, global dry z-score, 1-, 3-, and 6-hour relative differences, 24-to-168-hour standard deviation ratio, and ambient dose rate z-score and dose-to-gross ratio.

2. **Spectral Energy Shares and Pairwise Ratios (Tier 2, 30 features):** Normalized channel shares share_r0k = R0k/Gross for $k \in \{2, \ldots, 9\}$, high-energy share (R07+R08)/Gross, pairwise photopeak ratios (R05/R07, R03/R07, R05/R03, R02/R03), 6-hour rolling means, and 3-hour differences:

$$\text{share_high_energy} = \frac{\text{R07} + \text{R08}}{\text{Gross}}, \quad \text{ratio_r05_r07} = \frac{\text{R05}}{\text{R07} + 10^{-4}}, \quad \text{ratio_r03_r07} = \frac{\text{R03}}{\text{R07} + 10^{-4}} \tag{3}$$

3. **Multi-Scale Weather Fusion (Tier 3, 48 features):** Combines Tier 2 with the 15 NOAA weather features (1, 3, 6, and 24-hour precipitation, rain indicators, `hours_since_rain`, pressure and 3h/24h pressure changes, temperature, dew point, relative humidity) and 3 cross-domain interactions (`rain_high_energy_interaction`, `dry_excess_interaction`, `washout_expected_ratio`).

4. **Physics-Motivated Nuclear Features (Tier 4, 59 features):** Adds 11 features: global-dry and local-168h non-negative spectral stripping residuals projecting channel excesses onto fit-fold station radon basis vectors (`radon_stripping_residual_l2`, `stripped_excess_r03_z`, `stripped_excess_r05_z`, `stripped_excess_r07_z`, and local equivalents), channel-weighted dose consistency (`dose_spectral_residual_z`), and 3-stage Bateman washout ODE residuals (`bateman_washout_residual_ratio`, `post_rain_persistence_anomaly`). Both the scavenging exponent $\beta = 0.65$ in `solve_bateman_washout_ode` and the effective clearance lifetime $\tau_{\text{eff}} = 0.65$ h in `post_rain_persistence_anomaly` are hand-set assumed constants rather than fitted parameters.

Feature importance analysis following explainable AI evaluation frameworks for gamma-ray detection (Bandstra et al., 2023) shows that `hours_since_rain` is the single most influential predictor with an importance gain of 822,621, followed by `local_radon_stripping_residual_l2` (gain 629,970) and `z_score_global_dry` (gain 535,931). By contrast, `bateman_washout_residual_ratio` (gain 51,094, rank 14/59), `dose_spectral_residual_z` (gain 17,702, rank 21/59), and `post_rain_persistence_anomaly` (gain 856, rank 55/59) contribute much smaller split gains.

3.5 Synthetic Release Injections, Derived Detector Response, and Artifact Audit

Because the 2017–2025 station series contain no documented radiological releases, detection sensitivity is evaluated on 450 synthetic release plumes (250 train, 200 test) covering ^{137}Cs, ^{131}I, ^{134}Cs, ^{60}Co, and ^{132}Te/^{132}I across `linear_ramp` and `step` profiles (training) and `sigmoidal` and `exponential` profiles (testing), with active passage durations (`t_passage_hours`) of 3 to 28 hours and total durations (`duration_hours`) of 8 to 164 hours. Prior work has used atmospheric transport models (Stohl et al., 2012) and simulated gamma-ray response datasets (Buckton et al., 2025; Fjeldsted et al., 2024).

In the legacy v1 catalog (`synthetic_injection_catalog_legacy_v1.csv`), both training and test plumes drew channel shares from hand-assumed uniform/Dirichlet distributions with $\pm 15\%$ perturbations, added smooth noiseless curves, shared the station radon regression slope (`k_dose_channel`) for injected dose rates, and terminated 100% of plumes at a detected filter-tape drop. In our benchmark realism overhaul (`src/detector_response.py`, `src/artifact_tests.py`), we replaced this

generator with two independent physics-derived response families for the 2"×2" NaI(Tl) spectrometer (uncalibrated standoff, filter radius, housing thickness, and backscatter fraction documented in `response_templates.csv`): **Family A** (analytic disk-to-cylinder solid angle, NIST XCOM attenuation, Klein-Nishina Compton continuum, and Gaussian energy resolution broadening) for training plumes, and **Family B** (3D multi-scatter Monte Carlo photon ray-tracing with backscatter and gain/resolution perturbations) for test plumes. We also introduced Gamma-Poisson counting noise calibrated to empirical channel Fano factors (`artifact_counting_noise.csv`), an independent ICRP-74 + 3D detector efficiency dose model (`artifact_dose_circularity.csv`), and `continuous_arbitrary_end` plume timing where 62 of 200 test plumes cross an active filter change (dropping filter-age-only validation ROC-AUC from 0.6311 to 0.5195; `artifact_filter_age_leakage.csv`).

Stratifying by local 168-hour background SNR (`artifact_snr_stratification.csv`, Currie, 1968) shows that legacy sub-1σ gross plumes ($n = 18$, mean gross SNR 0.79σ) were detected 68.33% of the time on Tier 4 (65.28% on Tier 3) because concentrating 38–68% of counts into a single photopeak channel (R03/R05/R07, whose dry SD is $\sim 18\%$ of gross SD) produced smooth $> 2\sigma$ peak channel excesses. Under realistic Family A→B templates with counting noise and arbitrary-end timing, sub-1σ detection drops to 32.50% on Tier 2 (-41.67 pp), 36.67% on Tier 3 (-28.61 pp), and 60.83% on Tier 4 (-7.50 pp). Hold-out template experiments (`benchmark_holdout_templates.csv`, Lalor et al., 2026) further show that holding out ^{131}I (R03) reduces ^{131}I detection to 0.0%–1.67%, holding out ^{60}Co (R07) reduces detection to 0.0%–33.33%, and unseen test-only radionuclides peaking outside trained channels (^{103}Ru at 497 keV in R04: 0.5%–48.17%; ^{140}Ba/^{140}La at $537/1596$ keV in R04/R08: 0.0%–5.0%) fail while unseen ^{95}Zr/^{95}Nb (724–766 keV in R05) is detected at 96.0%–98.83%.

3.6 Multi-Tier Model Formulation, Calibration, and Operating-Threshold Selection

The machine learning framework utilizes gradient boosted decision trees (Ke et al., 2017) configured with a multi-class objective across 150 trees with maximum depth 6, num_leaves 31, learning rate 0.05, subsample 0.8, colsample_bytree 0.8, balanced class weights, and minimum child samples of 50. Models are fitted strictly on historical data from 2017 through 2020. Decision thresholds τ are selected on the held-out 2021–2022 validation split (74,199 valid hours = 8.46 station-years) and frozen prior to out-of-sample evaluation on the 2023–2025 test split (106,628 observed hours = 12.16 station-years).

Diagnosing probability saturation on the 2021–2022 validation fold (`docs/threshold_diagnosis.md`, `threshold_calibration_diagnostics.csv`) reveals that 0 of 74,199 clean validation hours (0.0%) have raw predicted fission probability $p_{\text{clean}} \geq 0.99$ for either Tier 3 or Tier 4 (only 3 hours for Tier 3 and 7 hours for Tier 4 exceed $p = 0.95$). Because the lowest threshold with 0 validation false-alarm episodes already achieves $\geq 95.06\%$ validation detection on Tier 2, Tier 3, and Tier 4, the 90% and 95% detection-target rules select the exact same threshold in 20 of 20 seeds on both the coarse `THRESHOLD_GRID` and finer `FINE_THRESHOLD_GRID` (`threshold_saturation_summary.csv`). To remove dependence on probability calibration and validation plumes, we also define operating points purely from quantiles of the clean-validation score distribution matching stated false-alarm budgets of $1, 2, 5, 10,$ and 20 validation FA/station-year (`benchmark_fa_budget_operating_points.csv`). On the realistic v2 benchmark across 20 seeds, Tier 4 achieves $98.20 \pm 0.44\%$ test detection (96.25% within ≤ 12 h; 12-hour time-normalized detection 97.60%) at 17.59 ± 3.41 test FA/yr at the 1 validation FA/yr budget ($\tau = 0.9436 \pm 0.0111$), and $95.40 \pm 0.85\%$ test detection (91.60% within ≤ 12 h) at 7.86 ± 1.10 test FA/yr at the zero-validation-FA ceiling ($\tau = 0.9848 \pm 0.0041$). Two-fold temporal cross-fitting of Platt and isotonic

calibrators inside 2021–2022 reduces Tier 4 hour-level Brier score by 73% (0.01155 → 0.00311) and event-level log loss by 96% (0.09350 → 0.00371) without altering rank-based FA-budget detection (`threshold_calibration_fa_budget_comparison.csv`). Paired vs. unpaired plume scoring differs by at most 0.5 pp (0.0 pp for Tier 4; `threshold_paired_vs_unpaired.csv`). Restricting positive training labels to active passage hours (1,536 vs. 4,952 hours; `threshold_label_window_sensitivity.csv`) cuts Tier 4 test false alarms at the 95% target from 8.55 to 2.71 FA/yr (92.0% detection), and upweighting rain hours 2× (`threshold_class_weight_sensitivity.csv`) halves test false alarms at the 5 FA/yr budget from 39.46 to 18.00 FA/yr (98.5% detection).

3.7 Benchmark Evaluation and Uncertainty Analysis

Benchmark performance across 20 random seeds on the 2023–2025 test dataset is summarized in Table 2.

Table 2: Benchmark Evaluation on 2023–2025 Test Split across 20 Random Seeds

Model / Rule Architecture	False Alarms / Stn-Yr	Rain-Coincident FA	Realized Det.	Median Delay
Standard Baseline (Rolling 7d 3σ)	82.62	32.5%	52.50%	10.0 h
Global Dry Baseline (Global $3\sigma_{\text{dry}}$)	52.86	24.9%	57.00%	13.5 h
Conservative Baseline (Rolling 7d 5σ)	23.51	57.6%	6.00%	8.0 h
Tier 1: Gross Radiation GBDT	102.83 ± 9.30	25.8%	$89.98 \pm 0.94\%$	6.1 h
Tier 1b: Gross Radiation + Weather GBDT	10.39 ± 2.62	0.0%	$79.62 \pm 1.47\%$	8.1 h
Tier 2: Spectral GBDT	27.97 ± 28.27	5.5%	$93.20 \pm 1.45\%$	4.0 h
Tier 3: Weather Fusion GBDT	3.89 ± 1.25	0.3%	$90.72 \pm 1.73\%$	4.0 h
Tier 4: Physics-Motivated Nuclear GBDT	$\mathbf{2.71 \pm 1.39}$	**1.2%**	$\mathbf{94.00 \pm 0.93\%}$	**4.4 h**
Tier 4: Physics-Motivated Nuclear (90% Target)	$\mathbf{2.63 \pm 1.49}$	**1.1%**	$\mathbf{93.82 \pm 1.24\%}$	**4.4 h**
Tier 3: Multi-Layer Perceptron	57.10 ± 23.67	3.5%	$89.62 \pm 2.93\%$	3.6 h

Station-month block bootstrap 95% confidence intervals for false alarm rates are [1.17, 4.58] for Tier 4 and [1.67, 5.41] for Tier 3, with overlapping intervals on the legacy v1 benchmark. Across 20 algorithmic seeds on identical data, Tier 4 produced lower false alarms in 12 of 20 seeds; adding the 11 Tier 4 features increases mean synthetic plume detection by +3.28 percentage points (94.00% vs. 90.72%) at the 95% validation target.

Following the pre-specified statistical analysis plan (`docs/analysis_plan.md`, `src/uncertainty.py`), primary inference on the realistic v2 benchmark replaces solver-seed standard deviations with **stratified plume cluster bootstrap** ($N = 200$ test plumes across 5×2 station-environment strata, $B = 10{,}000$) plus **exact Clopper-Pearson binomial intervals** for detection, and **station-month block bootstrap** ($N = 164$ observed station-months totaling 12.16 station-years, episode starts counted prior to block binning, $B = 10{,}000$) plus **exact Garwood Poisson intervals** for false-alarm rates (`uncertainty_single_detector_intervals.csv`). At the 95% validation detection target on seed 42, Tier 4 achieves 96.5% test detection (95% plume bootstrap CI $[94.0\%, 98.5\%]$; exact Clopper-Pearson $[92.9\%, 98.6\%]$) at 9.87 FA/yr (95% month-block bootstrap CI $[5.10, 15.97]$; exact Poisson $[8.18, 11.80]$; week-block bootstrap SE 1.72 vs. month-block SE 2.79, `uncertainty_week_vs_month_block_sensitivity.csv`), compared to Tier 3 at 93.5% detection ($[90.5\%, 96.0\%]$) and 12.99 FA/yr ($[7.04, 20.30]$), and non-ML `B4c` at 80.0% detection ($[75.0\%, 85.0\%]$) and 2.30 FA/yr ($[0.72, 4.69]$). In the pre-specified primary comparison (`Tier 4` vs. `B4c` at the 90% validation target, `uncertainty_paired_comparisons.csv`), paired bootstrap yields ΔDet = +26.5 pp ($[+21.0, +32.5$ pp$]$, $p = 0.0002$) with

ΔFA/yr $= +9.13$ ($[+4.50, +14.80]$, $p = 0.0002$); under full validation-and-test threshold resampling (`uncertainty_matched_operating_point_resampled.csv`), ΔDet $= +24.79$ pp ($[+13.5, +35.0$ pp$]$, $p = 0.0004$). For `Tier 4` vs. `Tier 3`, the false-alarm difference at the saturated 95% target includes zero (ΔFA/yr $= -3.12$, 95% CI $[-9.68, +3.18]$, $p_{\text{Holm}} = 0.6699$; threshold-resampled ΔDet $= +2.78$ pp $[-1.0, +6.5$ pp$]$, ΔFA/yr $= -19.34$ $[-96.06, +2.13]$), whereas at the 5 validation FA/yr clean-quantile budget `Tier 4` significantly reduces test false alarms relative to `Tier 3` under both fixed-threshold paired bootstrap (ΔFA/yr $= -49.08$, $[-73.63, -26.77]$, $p_{\text{Holm}} = 0.0004$) and full threshold resampling (ΔFA/yr $= -55.05$, $[-160.30, -16.46]$, $p = 0.0024$). Finally, a two-way random-effects ANOVA across 10 independent plume catalogs $\times$ 3 LightGBM seeds (`uncertainty_catalog_seed_variance.csv`) demonstrates that for Tier 4 at the 95% target, **plume-catalog variance accounts for 85.4% of total detection variance** ($\text{SD}_{\text{catalog}} = 3.72$ pp vs. $\text{SD}_{\text{seed}} = 1.53$ pp; grand mean 84.40%) and 71.7% **of false-alarm variance** ($\text{SD}_{\text{catalog}} = 2.23$ FA/yr vs. $\text{SD}_{\text{seed}} = 1.40$ FA/yr; 91.7% at the 5 FA/yr budget). Monte Carlo coverage verification across 500 synthetic datasets (`uncertainty_coverage_simulation.csv`) confirms empirical 95% coverage between 94.2% and 97.4%.

Because generator-artifact tests moved test detection by $>$ 2 pp on multiple model tiers, Table 3 reports the legacy v1 benchmark and the realistic v2 benchmark (`benchmark_realism_comparison.csv`) side by side across 20 random seeds.

Table 3: Side-by-Side Comparison: Legacy (v1) vs. Realistic (v2) Benchmark Across 20 Random Seeds (95% Validation Target)

Model Tier	Legacy v1 Det. (%)	Realistic v2 Det. (%)	Δ Det. (pp)	Legacy v1 FA/yr	Realistic v2 FA/yr	Realistic Delay
Tier 1 (Gross)	90.08 ± 1.33	87.95 ± 2.38	-2.13	104.77 ± 14.42	206.47 ± 52.91	8.20 h
Tier 1b (Gross+Weather)	79.28 ± 0.84	83.42 ± 1.60	$+4.14$	9.70 ± 1.69	55.43 ± 6.91	10.15 h
Tier 2 (Spectral)	93.53 ± 1.09	82.92 ± 3.60	-10.61	35.68 ± 41.01	0.33 ± 0.37	6.12 h
Tier 3 (Weather Fusion)	90.20 ± 1.72	85.83 ± 2.48	-4.37	3.44 ± 0.73	1.23 ± 1.08	5.90 h
Tier 4 (Physics-Motivated Nuclear)	$\mathbf{94.10 \pm 1.12}$	$\mathbf{93.60 \pm 1.03}$	$\mathbf{-0.50}$	$\mathbf{2.78 \pm 1.55}$	$\mathbf{7.19 \pm 1.17}$	**5.00** h



Figure 2: Receiver Operating Characteristic matched detection curves comparing baseline rules against machine learning tiers on the test split.

3.8 Leave-One-Station-Out Spatial Generalization

To assess spatial generalizability across distinct climates without local fine-tuning, models were evaluated in a leave-one-station-out cross-validation framework where each station was held out completely during training and threshold calibration. Results are reported in Table 4.

Table 4: Leave-One-Station-Out Spatial Generalization across Five Climate Regimes

Held-Out Station	Climate Regime	Tier 3 Det.	Tier 4 Det.	Tier 3 FA/yr	Tier 4 FA/yr
San Diego, CA	Coastal Mediterranean	70.0%	80.0%	0.00	1.40
Birmingham, AL	Humid Subtropical	85.0%	95.0%	0.00	0.00
Dallas, TX	Continental Convective	92.5%	95.0%	0.57	0.00
Washington, DC	Mid-Atlantic Temperate	95.0%	95.0%	1.18	0.00
Tampa, FL	Subtropical Convective	90.0%	97.5%	1151.30	169.42

Across Birmingham, Dallas, Washington, and San Diego, Tier 4 achieves ≤ 1.40 false alarms per station-year while detecting 80.0% to 95.0% of synthetic releases. In Tampa, however, unadapted zero-shot transfer yields 169.42 false alarms per year for Tier 4 (compared to

1,151.30 false alarms per year for Tier 3). Direct episode-level Tree SHAP diagnosis and multi-station spectral audits (`loso_tampa_false_alarms.csv`, `tampa_channel_share_comparison.csv`, `all_stations_yearly_baseline_drift.csv`) show that Tampa's clean 2023–2025 spectrum is a stark outlier (share_r02 $= 0.3376 \pm 0.0741$, share_r05 $= 0.0610 \pm 0.0075$, ratio_r05_r03 $= 0.1813 \pm 0.0072$ vs. share_r02 $\in [0.4592, 0.5086]$ and share_r05 $\in [0.0426, 0.0488]$ across 9 continental stations) after drifting upward from share_r05 $= 0.0430$ in 2017 to 0.0655 in 2023. On the v2 realistic benchmark (`seed=42`, 90% validation target), uncalibrated held-out Tampa produces 786.86 FA/yr ($1{,}751$ episodes, modal top-1 SHAP feature `share_r07_roll_6h`) for Tier 3 and 123.13 FA/yr (274 episodes, modal top-1 SHAP feature `ratio_r05_r03`) for Tier 4; counterfactually aligning only Tampa's static spectral shares/ratios using 2017–2020 dry stats cuts Tier 4 false alarms by 81% ($123.13 \rightarrow 23.37$ FA/yr, 100.0% detection; `tampa_gain_drift_hypothesis_test.csv`). Replacing static spectral shares/ratios and global-dry features with causal 168-hour rolling local deviations (`local_168h_baseline`, requiring zero target-station historical calibration beyond 7 days; `loso_normalization_variants.csv`) reduces held-out Tampa false alarms to 13.48 FA/yr (95.0% detection) for Tier 4 and 0.00 FA/yr on all 4 other pilot stations, while station standardization (`station_standardized`) reduces Tampa false alarms to 0.90 FA/yr (92.5% detection). Direct local calibration with $N \in \{0, 3, 6, 12\}$ months of trailing pre-test (2022) clean background (`loso_local_calibration_results.csv`) reduces Tier 4 Tampa false alarms from 123.13 FA/yr ($N = 0$) to 10.34 FA/yr ($N = 3$ m, 95.0% det), 1.35 FA/yr ($N = 6$ m, 87.5% det), and 0.00 FA/yr ($N = 12$ m zero-FA ceiling, 82.5% det; or 4.94 FA/yr at 97.5% det under a 5 FA/yr local budget). Across all 11 pre-registered stations (`extended_loso_11station_summary.csv`), zero-shot causal 168-hour local normalization (`ZeroShot_Local168h_Normalized`) keeps Tier 4 false alarms within 0.00–27.86 FA/yr across every station (including reducing volcanic basaltic Honolulu from 796.51 to 13.17 FA/yr at 82.5% detection).

3.9 The Rain-Masked Baseline Control Experiment

To test whether a simple heuristic muting gross-count alarms during rainfall suffices, a control experiment was evaluated across varying holdoff windows H . Results at matched detection targets are summarized in Table 5.

Table 5: Rain-Masked Baseline Control Experiment at Matched Detection Targets

Baseline / Mask Rule	Lowest FA/yr at $\geq 87.12\%$ Det.	Lowest FA/yr at $\geq 91.88\%$ Det.	Max Det.
Unmasked ($H =$ none)	215.48 ($k = 1.40$)	303.03 ($k = 0.75$)	92.50%
Rain Mask: Rain only ($H = 0$h)	197.47 ($k = 1.35$)	285.77 ($k = 0.75$)	92.50%
Rain Mask: Rain + 3h ($H = 3$h)	179.80 ($k = 1.30$)	251.65 ($k = 0.75$)	92.50%
Rain Mask: Rain + 6h ($H = 6$h)	181.77 ($k = 1.25$)	245.98 ($k = 0.75$)	92.50%
Rain Mask: Rain + 12h ($H = 12$h)	177.49 ($k = 1.25$)	238.66 ($k = 0.75$)	92.00%
Tier 3: Weather Fusion GBDT	3.89	3.89	96.50%
Tier 4: Physics-Motivated Nuclear GBDT	**2.63**	**2.71**	**98.00%**

The control experiment shows that suppressing gross-count alarms during rain barely reduces false alarms at high detection targets: to reach $\geq 91.88\%$ detection with gross counts alone, the threshold drops to $k = 0.75\sigma$, triggering 238.66 false alarms per year during dry hours.

To evaluate stronger non-ML, subspace, unsupervised, and simple supervised comparators on the realistic v2 benchmark without synthetic-plume leakage, we implemented eight baseline families (B1–B8 in `src/baselines.py`) fitted strictly on 2017–2020 clean background (B1–B6, B8) or 2017–2020 training plumes (B7) and thresholded on 2021–2022 validation (`baseline_matched_detection_table.csv`,

baseline_matched_fa_table.csv, Table 6). Among all detectors that never see synthetic plumes in training, **B4c** (Radon Projection + Stripped Channel Peak Z) is strongest: at the 5 and 10 validation FA/yr budgets, B4c achieves 89.0% test detection (81.5% within $\leq$ 12 h) at 6.49 false alarms per station-year (0.0% rain-coincident), and at the 95% validation target it achieves 80.0% test detection at 2.30 FA/yr. Thus, non-ML local spectral radon stripping (B4c) accounts for most of the false-alarm reduction over gross-count sigma rules ($227.07 \rightarrow 6.49$ FA/yr), while supervised GBDT training on synthetic templates (Tier 4: 96.5% test detection at 9.87 FA/yr on seed 42; 20-seed mean $93.60 \pm 1.03\%$ at 7.19 ± 1.17 FA/yr) contributes the remaining $+4.6$ to $+7.5$ pp detection gain.

Table 6: Stronger Baselines (B1–B8) and ML Feature Ablations on the Realistic v2 Benchmark (2017–2020 Fit, 2021–2022 Validation Threshold, 2023–2025 Test)

Detector Architecture	Plumes in Fit?	90% Val Target: Det / FA/yr	95% Val Target: Det / FA/yr	10 Val FA/yr Budget: Det / FA/yr
B1a: Rolling 168h Gross Z-Score	No	84.0% / 227.07	86.5% / 254.61	5.0% / 11.18
B2a: Rain-Masked Dry Z ($H = 6$h)	No	79.5% / 118.14*	79.5% / 118.14*	19.5% / 5.01
B3b: Weather Regression (Spline GAM)	No	65.5% / 72.26	76.5% / 101.45*	19.0% / 8.55
B4b: Radon Projection (Local 168h Norm)	No	77.0% / 47.35	82.0% / 59.36	31.0% / 14.88
B4c: Radon Projection + Peak Channel Z (Non-ML)	No	**70.0%** / **0.74**	**80.0%** / **2.30**	**89.0%** / **6.49**
B5b: Background NMF (2-Component Subspace)	No	79.5% / 129.89	80.0% / 276.80*	57.0% / 110.74
B6c: Unsupervised One-Class SVM (Tier 4, Bg Only)	No	86.0% / 103.01	78.5% / 41.43	72.0% / 9.29
B7a: Supervised Logistic Regression (Tier 3)	Yes	61.5% / 24.42	70.5% / 23.59	71.5% / 24.17
B8: Operational Approx. (approximate EPA Threshold)	No	62.0% / 203.47*	62.0% / 203.47*	11.0% / 7.81
Ablation: Tier 4 without Weather/Bateman (39 feats)	Yes	93.0% / 6.33	93.0% / 6.33	93.0% / 6.33
Ablation: Tier 3 + Radon Stripping Only (57 feats)	Yes	95.5% / 7.48	95.5% / 7.48	95.5% / 7.48
Tier 4: Physics-Motivated Nuclear GBDT (Seed 42)	Yes	**96.5%** / **9.87**	**96.5%** / **9.87**	**96.5%** / **9.87**

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